

DATASET DISCOVERY & AI RESOURCE PLATFORM

Project Overview

Curated dataset discovery and management portal connecting data engineers and machine learning practitioners with structured datasets across CV, NLP, and tabular domains.

Problem Statement:	ML developers spend excessive time searching for clean, verified, properly formatted datasets distributed across scattered web sources.
Solution Implemented:	Built a high-performance dataset catalog with real-time text search, category filtering, schema previews, and direct access links.
My Technical Role:	Lead Full-Stack Web Developer

Main Features & Technical Stack

Core Features:	Technologies Used:
- Categorized directory (Computer Vision, NLP, Tabular, Audio) - Instant client-side search with dynamic tag filter - Dataset detail cards featuring sample rows & schema specs - Developer dataset upload workflow with format verification	React, Vite, JavaScript, HTML5, CSS3 Glassmorphism, Node.js API, Vercel
	Architecture Flow:
	React Interface → REST API Router → Search Indexing Engine → Metadata Store → Static File CDN

Technical Challenge & Results

Key Engineering Challenge: Optimized catalog search indexing to maintain sub-16ms UI frame rates on mobile viewports during fast typing.

Validated Results: Accelerated dataset discovery for ML developers with instant category filtering.